

How Nearly Repeating Parity Forces a Return: A Friendly Guide to Finite Collatz Defects

Companion to “Finite Parity Defects Force Collatz Returns”*

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1 The result in ordinary language

Use the shortcut Collatz rule: halve an even number; triple an odd number, add one, and halve it. Record a 1 at each odd number and a 0 at each even number. Now compare this parity diary with a copy shifted forward by q places.

If the two copies disagree only a few times over a sufficiently long finite stretch, the orbit must actually return to an earlier number. The theorem gives both the required stretch length and a short list of checkpoints where the return must occur.

Let K satisfy $N + 1 \leq 2^K$ for the starting number N . For instance, $N = 3$ allows $K = 2$. If there are at most D disagreements, the sufficient number of comparisons is

$$L = (2^{D+1} - 1)(q + K).$$

Those comparisons use $L + q$ parity letters. The possible return checkpoints are

$$s_k = (2^k - 1)(q + K), \quad k = 0, 1, \dots, D.$$

What Lean proves. At one of these checkpoints, the orbit value equals the value q steps later. For $q > 0$, the future repeats and the orbit is bounded. For $q = 2$ and a positive starting number, the theorem goes further: the orbit reaches one.

The result allows disagreements anywhere in the checked interval. It does not assume a particular Sturmian slope or a favorable density of even steps. The cost is that the sufficient length grows exponentially with the allowed number of disagreements.

This is a new proved criterion in the repository. We have not proved that every positive starting number satisfies it.

*Written with Codex (OpenAI). The technical paper and Lean source give the exact statements and proofs. No claim of literature priority is made.

2 Why an exact interval matters

Two distinct starting numbers sharing their first r parity letters differ by a nonzero multiple of 2^r . Their difference is therefore at least 2^r . This is the arithmetic fact behind the earlier exact repetition criterion.

There is also a simple size bound. After u shortcut steps,

$$T^u(N) + 1 \leq 2^u(N + 1) \leq 2^{u+K}.$$

It is deliberately generous, but holds for every choice of parities.

At a time s , compare the two orbit values $T^s(N)$ and $T^{s+q}(N)$. If their parity diaries agree for the next $s + q + K$ letters, the divisibility lower bound is too large for these two values to be distinct. They must be equal.

The local choice. At every checkpoint s , either the orbit returns after q steps, or there is a parity disagreement before time $2s + q + K$.

Set $A = q + K$. Choose checkpoints $0, A, 3A, 7A, \dots$ so that the search windows fit together without overlapping:

Checkpoint	Window in which a defect is forced	Next checkpoint
0	$[0, A)$	A
A	$[A, 3A)$	$3A$
$3A$	$[3A, 7A)$	$7A$
$7A$	$[7A, 15A)$	$15A$

If every one of $D + 1$ checkpoints failed to return, these disjoint windows would contain $D + 1$ different disagreements. That contradicts the allowed total of D .

This replaces a direct attempt to bound swap costs. Moving one parity letter across a late adjacent position can have a very large weighted arithmetic cost. The window argument instead uses the exact stretches between disagreements.

3 Allowing edits to a repeating template

Suppose a finite parity diary differs in at most e letters from a repeating template with period q . Each edited letter can affect at most two comparisons with the shifted copy: one in which it is the first letter, and one in which it is the second.

Thus at most e edited letters cause at most $2e$ shift disagreements. The sufficient comparison length becomes

$$L = (2^{2e+1} - 1)(q + K),$$

and the template is checked for $L + q$ letters. The edit positions can be arbitrary. Lean checks this extension too.

For a concrete example, start at 3:

$$3, 5, 8, 4, 2, 1, 2, 1, \dots$$

Its diary begins 11000101... Compare it with its shift by two places. The only disagreements are at times 0, 1, 3. With $D = 3$, $q = 2$, and $K = 2$, the sufficient comparison length is 60. The checkpoints are 0, 4, 12, 28. Already at time 4, the value 2 equals the value two steps later.

The same diary differs from the template 010101... only at times 0 and 3. This gives $e = 2$ edited letters, whose more conservative template bound is $L = 124$. The distinction matters: edited letters and shift disagreements are different counts.

4 What remains between this and Collatz

The missing step. We know what happens if a finite parity prefix is sufficiently close to periodic. We do not know that every hypothetical counterexample must have such a prefix.

For a general positive period, the return could belong to a nontrivial cycle. For period two, Lean proves that the only positive return values are 1 and 2, so the certificate implies convergence to one. But its universal availability is still unproved. Every orbit reaching one eventually alternates; assuming that all orbits have the needed period-two certificate would simply rephrase the convergence problem.

The next useful advance needs an independent restriction on the number of defects, or a sharper bound on the time available between defects. The expanded audit checks 35 selected declarations using only Lean's standard foundational axioms; it is not an independent kernel replay.

The technical paper is `paper/defects.pdf`, with source `paper/defects.tex`. The formal development is `lean/Collatz/ParityDefects.lean`; the research note is `analysis/PARITY_DEFECTS.md`.